Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c,\ b = d$$

Powers

$$i^4 = 1 \qquad i^{4q+r} = i^r$$

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Worked Solutions **Complex Numbers**

Q001 Written

State the real and imaginary parts of $3 - 2i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(3,\ -2)$.

Final answer

$(3,\ -2)$

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(4/3, -\sqrt{5}/3)$.

Final answer

$(4/3, -\sqrt{5}/3)$

Worked Solutions **Complex Numbers**

Q003 Written

State the real and imaginary parts of -8.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-8,\ 0)$.

Final answer

$(-8,\ 0)$

Q004 Written

State the real and imaginary parts of $2i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ 2)$.

Final answer

$(0,\ 2)$

Worked Solutions **Complex Numbers**

Q005 Written

Find the real values of x and y satisfying $(2+i)x+(1+i)y=2+3i, x, yreal$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=-1, y=4$.

Final answer

$x=-1, y=4$

Q006 Written

State the real and imaginary parts of $-3+5i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-3,\ 5)$.

Final answer

$(-3,\ 5)$

Worked Solutions **Complex Numbers**

Q007 **Written**

State the real and imaginary parts of $(-1-\sqrt{3}i)/2$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-1/2, -\sqrt{3}/2)$.

Final answer

$(-1/2, -\sqrt{3}/2)$

Q008 **Written**

State the real and imaginary parts of 1.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1,\ 0)$.

Final answer

$(1,\ 0)$

Worked Solutions **Complex Numbers**

Q009 Written

State the real and imaginary parts of $3i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ 3)$.

Final answer

$(0,\ 3)$

Q010 Written

Find the real values of x and y satisfying $(x-6)+(x+3y)i=0, x, yreal$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=6, y=-2$.

Final answer

$x=6, y=-2$

Worked Solutions **Complex Numbers**

Q011 **Written**

Find the real values of x and y satisfying $(3+2i)x+(4-i)y=6-7i, x, yreal$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=-2, y=3$.

Final answer

$x=-2, y=3$

Q012 **Written**

State the real and imaginary parts of $2+5i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2,\ 5)$.

Final answer

$(2,\ 5)$

Worked Solutions **Complex Numbers**

Q013 Written

State the real and imaginary parts of $-25 + 13i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-25,\ 13)$.

Final answer

$(-25,\ 13)$

Q014 Written

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2/3, -\sqrt{5}/3)$.

Final answer

$(2/3, -\sqrt{5}/3)$

Worked Solutions **Complex Numbers**

Q015 **Written**

State the real and imaginary parts of $(7-\sqrt{2}i)/2$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(7/2, -\sqrt{2}/2)$.

Final answer

$(7/2, -\sqrt{2}/2)$

Q016 **Written**

State the real and imaginary parts of -4.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-4,\ 0)$.

Final answer

$(-4,\ 0)$

Worked Solutions **Complex Numbers**

Q017 Written

State the real and imaginary parts of $1/4$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1/4,\ 0)$.

Final answer

$(1/4,\ 0)$

Q018 Written

State the real and imaginary parts of $\sqrt{3}i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, \sqrt{3})$.

Final answer

$(0, \sqrt{3})$

Worked Solutions **Complex Numbers**

Q019 Written

State the real and imaginary parts of $-i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ -1)$.

Final answer

$(0,\ -1)$

Q020 Written

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = 2$.

Final answer

$x = 3, y = 2$

Worked Solutions **Complex Numbers**

Q021 **Written**

Find the real values of x and y satisfying $(x+1)-5i=4+(2y+1)i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=3, y=-3$.

Final answer

$x=3, y=-3$

Q022 **Written**

Find the real values of x and y satisfying $(x-2)+(x+y)i=0$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=2, y=-2$.

Final answer

$x=2, y=-2$

Worked Solutions **Complex Numbers**

Q023 **Written**

Find the real values of x and y satisfying $(3+2i)x+(1-i)y=7+3i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=2, y=1$.

Final answer

$x=2, y=1$

Q024 **Written**

Find the real values of x and y satisfying $(1-i)(x+yi)=2+i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=1/2, y=3/2$.

Final answer

$x=1/2, y=3/2$

Worked Solutions **Complex Numbers**

Q025 **Written**

Solve the complex-number equation and find the exact real values of x and y: $(1 + 2i)(x + i) = x(y + i)$.

Solution

1. Expand: $((1 + 2i)(x + i) = x - 2 + (1 + 2x)i)$.
2. Equate coefficients with (xy+xi): $(x - 2 = xy)$ and $(1 + 2x = x)$.
3. Solving gives $x = -1, y = 3$.

Final answer

$x = -1, y = 3$

Q026 **Written**

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 7i$.

Final answer

$1 - 7i$

Worked Solutions **Complex Numbers**

Q027 Written

Calculate $(i-2)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $3-4i$.

Final answer

$3-4i$

Q028 Written

Calculate $i^6+i^{27}+i^{17}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -1.

Final answer

-1

Worked Solutions **Complex Numbers**

Q029 Written

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 + 8i$.

Final answer

$1 + 8i$

Q030 Written

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $8 - i$.

Final answer

$8 - i$

Worked Solutions **Complex Numbers**

Q031 Written

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 0.

Final answer

0

Q032 Written

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $4 - i$.

Final answer

$4 - i$

Worked Solutions **Complex Numbers**

Q033 Written

Calculate $(-4+3i)+(5-6i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1-3i$.

Final answer

$1-3i$

Q034 Written

Calculate $(3-3i)-(5-3i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -2.

Final answer

-2

Worked Solutions **Complex Numbers**

Q035 **Written**

Calculate $(\sqrt{2}+3i)(\sqrt{2}+i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-1+4\sqrt{2}i$.

Final answer

$-1+4\sqrt{2}i$

Q036 **Written**

Calculate $(2+3i)+(5+8i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $7+11i$.

Final answer

$7+11i$

Worked Solutions **Complex Numbers**

Q037 Written

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1/15 - 13/14i$.

Final answer

$1/15 - 13/14i$

Q038 Written

Calculate $(2i - 1)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-3 - 4i$.

Final answer

$-3 - 4i$

Worked Solutions **Complex Numbers**

Q039 Written

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -3.

Final answer

-3

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 4.

Final answer

4

Worked Solutions **Complex Numbers**

Q041 Written

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: i.

Final answer

i

Q042 Written

Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -1.

Final answer

-1

Worked Solutions **Complex Numbers**

Q043 **Written**

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-i$.

Final answer

$-i$

Q044 **Written**

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2 i^{40}$.

Solution

1. Use ($i^{40} = 1$) and (($x - i)(x + i) = x^2 + 1$).
2. The equation becomes ($7x(x^2 + 1) = 14x^2$), so ($7x(x - 1)^2 = 0$).
3. Therefore $x = 0$ or $x = 1$.

Final answer

$x = 0$ or $x = 1$

Worked Solutions **Complex Numbers**

Q045 Written

State the complex conjugate of $2 + 3i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $2 - 3i$.

Final answer

$2 - 3i$

Q046 Written

State the complex conjugate of $7 - i\sqrt{11}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $7 + i\sqrt{11}$.

Final answer

$7 + i\sqrt{11}$

Worked Solutions **Complex Numbers**

Q047 **Written**

State the complex conjugate of 6.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 6.

Final answer

6

Q048 **Written**

State the complex conjugate of $-5i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $5i$.

Final answer

$5i$

Worked Solutions **Complex Numbers**

Q049 **Written**

Calculate $(2+i)/(2-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$3/5 + 4/5i$

Q050 **Written**

Calculate $(3-i)/(2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1/2 - 3/2i$.

Final answer

$-1/2 - 3/2i$

Q051 Written

State the complex conjugate of $1 - 4i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $1 + 4i$.

Final answer

$1 + 4i$

Q052 Written

State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{2} - i\sqrt{10}$.

Final answer

$\sqrt{2} - i\sqrt{10}$

Worked Solutions **Complex Numbers**

Q053 **Written**

State the complex conjugate of 2.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 2.

Final answer

2

Q054 **Written**

State the complex conjugate of $-8i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8i$.

Final answer

$8i$

Worked Solutions **Complex Numbers**

Q055 **Written**

Calculate $(3+i)/(1+2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1-i$.

Final answer

$1-i$

Q056 **Written**

Calculate $(1-i)/i$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1-i$.

Final answer

$-1-i$

Worked Solutions **Complex Numbers**

Q057 Written

State the complex conjugate of $3 + 4i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $3 - 4i$.

Final answer

$3 - 4i$

Q058 Written

State the complex conjugate of $9 + 8i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $9 - 8i$.

Final answer

$9 - 8i$

Worked Solutions **Complex Numbers**

Q059 **Written**

State the complex conjugate of $8 + i\sqrt{3}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8 - i\sqrt{3}$.

Final answer

$8 - i\sqrt{3}$

Q060 **Written**

State the complex conjugate of 5.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 5.

Final answer

5

Worked Solutions **Complex Numbers**

Q061 **Written**

State the complex conjugate of -15.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is -15.

Final answer

-15

Q062 **Written**

State the complex conjugate of $\sqrt{7}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{7}$.

Final answer

$\sqrt{7}$

Worked Solutions **Complex Numbers**

Q063 **Written**

State the complex conjugate of i.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i$.

Final answer

$-i$

Q064 **Written**

State the complex conjugate of $-50i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $50i$.

Final answer

$50i$

Worked Solutions **Complex Numbers**

Q065 Written

State the complex conjugate of $i\sqrt{6}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i\sqrt{6}$.

Final answer

$-i\sqrt{6}$

Q066 Written

Calculate $(1+i)/(1-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is i.

Final answer

i

Worked Solutions **Complex Numbers**

Q067 **Written**

Calculate $(3+i)/(2+i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $7/5 - 1/5i$.

Final answer

$7/5 - 1/5i$

Q068 **Written**

Calculate $(\sqrt{5}+i)/(\sqrt{5}-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2/3 + \sqrt{5}/3i$.

Final answer

$2/3 + \sqrt{5}/3i$

Worked Solutions **Complex Numbers**

Q069 Written

Calculate $5i/(1+2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2+i$.

Final answer

$2+i$

Q070 Written

Calculate $5i/(4+3i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5+4/5i$.

Final answer

$3/5+4/5i$

Worked Solutions **Complex Numbers**

Q071 Written

Calculate $(1-3i)/(3-2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $9/13-7/13i$.

Final answer

$9/13-7/13i$

Q072 Written

Calculate $1/i$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-i$.

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q073 **Written**

Calculate $(2 + i)/(3i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1/3 - 2/3i$.

Final answer

$1/3 - 2/3i$

Q074 **Written**

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)}\right)^{2027}$.

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The base simplifies exactly to (displaystyle
rac-2-144i85).
3. Hence the exact result is $((-2 - 144i)/85)^{2027}$; no expansion of the 2027th power is required.

Final answer

$((-2 - 144i)/85)^{2027}$

Worked Solutions **Complex Numbers**

Q075 Written

Express the square roots of $-6intermsofi$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm i\sqrt{6}$.

Final answer

$\pm i\sqrt{6}$

Q076 Written

Express $\sqrt{-12}\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-6\sqrt{2}$.

Final answer

$-6\sqrt{2}$

Worked Solutions **Complex Numbers**

Q077 **Written**

Express $\sqrt{8}/\sqrt{-18}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-(2/3)i$.

Final answer

$-(2/3)i$

Q078 **Written**

Express $\sqrt{-7}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $i\sqrt{7}$.

Final answer

$i\sqrt{7}$

Worked Solutions **Complex Numbers**

Q079 Written

Express $\sqrt{-27}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i\sqrt{3}$.

Final answer

$3i\sqrt{3}$

Q080 Written

Express $-\sqrt{-12}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i\sqrt{3}$.

Final answer

$-2i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q081 Written

Express the square roots of -12 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 2i\sqrt{3}$.

Final answer

$\pm 2i\sqrt{3}$

Q082 Written

Express $\sqrt{-3}\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-3\sqrt{2}$.

Final answer

$-3\sqrt{2}$

Worked Solutions **Complex Numbers**

Q083 **Written**

Express $\sqrt{27}/\sqrt{-3}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-3i$.

Final answer

$-3i$

Q084 **Written**

Express $(3+\sqrt{-2})^2$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $7+6\sqrt{2}i$.

Final answer

$7+6\sqrt{2}i$

Worked Solutions **Complex Numbers**

Q085 Written

Express $\sqrt{-2}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $i\sqrt{2}$.

Final answer

$i\sqrt{2}$

Q086 Written

Express $\sqrt{-9}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i$.

Final answer

$3i$

Worked Solutions **Complex Numbers**

Q087 Written

Express $\sqrt{-45}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i\sqrt{5}$.

Final answer

$3i\sqrt{5}$

Q088 Written

Express $-\sqrt{-24}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i\sqrt{6}$.

Final answer

$-2i\sqrt{6}$

Worked Solutions **Complex Numbers**

Q089 Written

Express $-\sqrt{-36}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-6i$.

Final answer

$-6i$

Q090 Written

Express the square roots of -2 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm i\sqrt{2}$.

Final answer

$\pm i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q091 Written

Express the square roots of -18 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 3i\sqrt{2}$.

Final answer

$\pm 3i\sqrt{2}$

Q092 Written

Express the square roots of -16 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 4i$.

Final answer

$\pm 4i$

Worked Solutions **Complex Numbers**

Q093 **Written**

Express $\sqrt{-2}\,\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2\sqrt{3}$.

Final answer

$-2\sqrt{3}$

Q094 **Written**

Express $\sqrt{-28}\,\sqrt{-35}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-14\sqrt{5}$.

Final answer

$-14\sqrt{5}$

Worked Solutions **Complex Numbers**

Q095 **Written**

Express $\sqrt{8}/\sqrt{-2}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i$.

Final answer

$-2i$

Q096 **Written**

Express $\sqrt{27}/\sqrt{-12}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-(3/2)i$.

Final answer

$-(3/2)i$

Worked Solutions **Complex Numbers**

Q097 Written

Express $\sqrt{-32}+\sqrt{-98}+\sqrt{-8}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $13i\sqrt{2}$.

Final answer

$13i\sqrt{2}$

Q098 Written

Express $\sqrt{-98}-\sqrt{-72}+\sqrt{-50}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $6i\sqrt{2}$.

Final answer

$6i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q099 **Written**

Express $(\sqrt{3}+\sqrt{-5})^2$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2+2i\sqrt{15}$.

Final answer

$-2+2i\sqrt{15}$

Q100 **Written**

Express $(2+\sqrt{-3})(2-\sqrt{-3})$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore 7.

Final answer

7

Worked Solutions **Complex Numbers**

Q101 Written

Solve over the complex numbers: $(2x-8)^2 = -10$.

Solution

1. Take square roots: $(2x - 8 = pm$ isqrt10).
2. Divide by 2 to obtain $x = 4 \pm (\sqrt{10}/2)i$.

Final answer

$x = 4 \pm (\sqrt{10}/2)i$

Q102 MCQ

For $2 + 3i$, **select the ordered pair (real part, imaginary part):**

Solution

1. Simplify $2 + 3i$ using the relevant family rule.
2. The correct value is $(2,\ 3)$.

Correct option: C

Final answer

$(2,\ 3)$

Worked Solutions **Complex Numbers**

Q103 MCQ

For $-4+5i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-4+5i$ using the relevant family rule.
2. The correct value is $(-4,\ 5)$.

Correct option: D

Final answer

$(-4,\ 5)$

Q104 MCQ

For $7-2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $7-2i$ using the relevant family rule.
2. The correct value is $(7,\ -2)$.

Correct option: D

Final answer

$(7,\ -2)$

Worked Solutions **Complex Numbers**

Q105 MCQ

For $(1-\sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $(1-\sqrt{2}i)/3$ using the relevant family rule.
2. The correct value is $\left(\frac{1}{3},\ -\frac{\sqrt{2}}{3}\right)$.

Correct option: B

Final answer

$\left(\frac{1}{3},\ -\frac{\sqrt{2}}{3}\right)$

Q106 MCQ

For -6, select the ordered pair (real part, imaginary part):

Solution

1. Simplify -6 using the relevant family rule.
2. The correct value is $(-6,\ 0)$.

Correct option: D

Final answer

$(-6,\ 0)$

Worked Solutions **Complex Numbers**

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4i$ using the relevant family rule.
2. The correct value is $(0,\ 4)$.

Correct option: A

Final answer

$(0,\ 4)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-i$ using the relevant family rule.
2. The correct value is $(0,\ -1)$.

Correct option: B

Final answer

$(0,\ -1)$

Worked Solutions **Complex Numbers**

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{5}i$ using the relevant family rule.
2. The correct value is $\left(0,\ \sqrt{5}\right)$.

Correct option: C

Final answer

$\left(0,\ \sqrt{5}\right)$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $3/2 + \sqrt{3}i$ using the relevant family rule.
2. The correct value is $\left(\frac{3}{2},\ \sqrt{3}\right)$.

Correct option: B

Final answer

$\left(\frac{3}{2},\ \sqrt{3}\right)$

Worked Solutions **Complex Numbers**

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-1/4 - 2i$ using the relevant family rule.
2. The correct value is $\left(-\frac{1}{4},\ -2\right)$.

Correct option: D

Final answer

$\left(-\frac{1}{4},\ -2\right)$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $5/3$ using the relevant family rule.
2. The correct value is $\left(\frac{5}{3},\ 0\right)$.

Correct option: D

Final answer

$\left(\frac{5}{3},\ 0\right)$

Worked Solutions **Complex Numbers**

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{7}i$ using the relevant family rule.
2. The correct value is $\left(0,\ -\sqrt{7}\right)$.

Correct option: C

Final answer

$\left(0,\ -\sqrt{7}\right)$

Q114 MCQ

For $-9+\sqrt{2}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-9+\sqrt{2}i$ using the relevant family rule.
2. The correct value is $\left(-9,\ \sqrt{2}\right)$.

Correct option: D

Final answer

$\left(-9,\ \sqrt{2}\right)$

Worked Solutions **Complex Numbers**

Q115 *MCQ*

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2/5 - 3/4i$ using the relevant family rule.
2. The correct value is $\left(\frac{2}{5}, -\frac{3}{4}\right)$.

Correct option: B

Final answer

$\left(\frac{2}{5}, -\frac{3}{4}\right)$

Q116 *MCQ*

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{11}$ using the relevant family rule.
2. The correct value is $\left(-\sqrt{11},\ 0\right)$.

Correct option: B

Final answer

$\left(-\sqrt{11},\ 0\right)$

Worked Solutions **Complex Numbers**

Q117 *MCQ*

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{13}i$ using the relevant family rule.

2. The correct value is $\left(0,\ \sqrt{13}\right)$.

Correct option: C

Final answer

$\left(0,\ \sqrt{13}\right)$

Q118 *MCQ*

For $(2+i)x+(1-i)y=3+3i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.

2. Solving the real system gives $(2,\ -1)$.

Correct option: D

Final answer

$(2,\ -1)$

Worked Solutions **Complex Numbers**

Q119 MCQ

For $(1+2i)x+(3-i)y=5-4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1,\ 2)$.

Correct option: A

Final answer

$(-1,\ 2)$

Q120 MCQ

For $(3-2i)x+(1+i)y=6+i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1,\ 3)$.

Correct option: B

Final answer

$(1,\ 3)$

Worked Solutions **Complex Numbers**

Q121 *MCQ*

For $(2-i)x+(4+2i)y=4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-2,\ 1)$.

Correct option: B

Final answer

$(-2,\ 1)$

Q122 *MCQ*

For $(1+i)x+(2+3i)y=1$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(3,\ -1)$.

Correct option: C

Final answer

$(3,\ -1)$

Worked Solutions **Complex Numbers**

Q123 MCQ

For $(4+i)x+(1-2i)y=2+5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1,\ -2)$.

Correct option: D

Final answer

$(1,\ -2)$

Q124 MCQ

For $(2+3i)x+(3+i)y=-8-5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1,\ -2)$.

Correct option: D

Final answer

$(-1,\ -2)$

Worked Solutions **Complex Numbers**

Q125 MCQ

For $(1-i)x+(2-i)y=6-4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2,\ 2)$.

Correct option: C

Final answer

$(2,\ 2)$

Q126 MCQ

For $(3+4i)-(1-2i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $2+6i$.

Correct option: C

Final answer

$2+6i$

Worked Solutions **Complex Numbers**

Q127 MCQ

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2 = -1)$, reducing powers of i modulo 4.
2. The simplified value is $3 - 2i$.

Correct option: C

Final answer

$3 - 2i$

Q128 MCQ

For $(1 + 2i)(3 - i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2 = -1)$, reducing powers of i modulo 4.
2. The simplified value is $5 + 5i$.

Correct option: D

Final answer

$5 + 5i$

Worked Solutions **Complex Numbers**

Q129 *MCQ*

For $(\sqrt{3}+i)(\sqrt{3}-i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 4.

Correct option: A

Final answer

4

Q130 *MCQ*

For $(2-i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2-i)^2$.

Correct option: C

Final answer

$(2-i)^2$

Worked Solutions **Complex Numbers**

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-8 + 6i$.

Correct option: A

Final answer

$-8 + 6i$

Q132 MCQ

For i^{37}, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is i.

Correct option: B

Final answer

i

Worked Solutions **Complex Numbers**

Q133 MCQ

For $i^{22} + i^9$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: B

Final answer

$-1 + i$

Q134 MCQ

For $i^{15} - i^6$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $1 - i$.

Correct option: B

Final answer

$1 - i$

Worked Solutions **Complex Numbers**

Q135 MCQ

For $(4-2i)+(1+5i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $5+3i$.

Correct option: C

Final answer

$5+3i$

Q136 MCQ

For $(2+i)(2-i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is 5.

Correct option: A

Final answer

5

Worked Solutions **Complex Numbers**

Q137 MCQ

For $(\sqrt{2}+2i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $\left(\sqrt{2}+2i\right)^2$.

Correct option: C

Final answer

$\left(\sqrt{2}+2i\right)^2$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-i$.

Correct option: C

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q139 MCQ

For $(3-i)^3$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(3-i)^3$.

Correct option: D

Final answer

$(3-i)^3$

Q140 MCQ

For $(1-2i)(-2+i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5i$.

Correct option: D

Final answer

$5i$

Worked Solutions **Complex Numbers**

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: D

Final answer

$-1 + i$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: D

Final answer

$2 + 6i$

Worked Solutions **Complex Numbers**

Q143 MCQ

For $(2-3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2-3i)^2$.

Correct option: C

Final answer

$(2-3i)^2$

Q144 MCQ

For the expression State the complex conjugate of $3-2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $3+2i$.

Correct option: C

Final answer

$3+2i$

Worked Solutions **Complex Numbers**

Q145 *MCQ*

For the expression State the complex conjugate of $-4+\sqrt{3}i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-4-\sqrt{3}i$.

Correct option: A

Final answer

$-4-\sqrt{3}i$

Q146 *MCQ*

For the expression State the complex conjugate of $7i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-7i$.

Correct option: C

Final answer

$-7i$

Worked Solutions **Complex Numbers**

Q147 **MCQ**

For the expression State the complex conjugate of $-\sqrt{5}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\sqrt{5}$.

Correct option: C

Final answer

$-\sqrt{5}$

Q148 **MCQ**

For the expression State the complex conjugate of $1/2 - 3i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + 3i$.

Correct option: B

Final answer

$\frac{1}{2} + 3i$

Worked Solutions **Complex Numbers**

Q149 MCQ

For $(2+i)/(1-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2}+\frac{3i}{2}$.

Correct option: A

Final answer

$\frac{1}{2}+\frac{3i}{2}$

Q150 MCQ

For $(4-3i)/(2+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $1-2i$.

Correct option: A

Final answer

$1-2i$

Worked Solutions **Complex Numbers**

Q151 MCQ

For $(1 + 2i)/(3 - i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{10} + \frac{7i}{10}$.

Correct option: B

Final answer

$\frac{1}{10} + \frac{7i}{10}$

Q152 MCQ

For $5i/(2 - i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $i\,(2 + i)$.

Correct option: C

Final answer

$i\,(2 + i)$

Worked Solutions **Complex Numbers**

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D

Final answer

$-i$

Q154 MCQ

For $1/i$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q155 MCQ

For $(2+3i)/(2-3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13}+\frac{12i}{13}$.

Correct option: B

Final answer

$-\frac{5}{13}+\frac{12i}{13}$

Q156 MCQ

For $(\sqrt{2}+i)/(\sqrt{2}-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{2}+i}{\sqrt{2}-i}$.

Correct option: B

Final answer

$\frac{\sqrt{2}+i}{\sqrt{2}-i}$

Worked Solutions **Complex Numbers**

Q157 MCQ

For $(1-4i)/(3+2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13}-\frac{14i}{13}$.

Correct option: D

Final answer
$-\frac{5}{13}-\frac{14i}{13}$

Q158 MCQ

For $7i/(1-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{7}{2}+\frac{7i}{2}$.

Correct option: C

Final answer
$-\frac{7}{2}+\frac{7i}{2}$

Worked Solutions **Complex Numbers**

Q159 MCQ

For the expression State the complex conjugate of $-2+5i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-2-5i$.

Correct option: B

Final answer

$-2-5i$

Q160 MCQ

For the expression State the complex conjugate of $6-\sqrt{7}i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $6+\sqrt{7}i$.

Correct option: D

Final answer

$6+\sqrt{7}i$

Worked Solutions **Complex Numbers**

Q161 *MCQ*

For the expression State the complex conjugate of $-i\sqrt{2}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{2}i$.

Correct option: B

Final answer

$\sqrt{2}i$

Q162 *MCQ*

For the expression State the complex conjugate of 9:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is 9.

Correct option: C

Final answer

9

Worked Solutions **Complex Numbers**

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{3}{4} - 2i$.

Correct option: A

Final answer

$-\frac{3}{4} - 2i$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{5} + \frac{12i}{5}$.

Correct option: D

Final answer

$\frac{1}{5} + \frac{12i}{5}$

Worked Solutions **Complex Numbers**

Q165 MCQ

For $(2-i)/(4+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{7}{17} - \frac{6i}{17}$.

Correct option: A

Final answer

$\frac{7}{17} - \frac{6i}{17}$

Q166 MCQ

For $3i/(2+3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{9}{13} + \frac{6i}{13}$.

Correct option: A

Final answer

$\frac{9}{13} + \frac{6i}{13}$

Worked Solutions **Complex Numbers**

Q167 MCQ

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{3} - i}{\sqrt{3} + i}$.

Correct option: C

Final answer

$$\frac{\sqrt{3} - i}{\sqrt{3} + i}$$

Q168 MCQ

For $(1 + i)/(2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} - \frac{i}{2}$.

Correct option: A

Final answer

$$\frac{1}{2} - \frac{i}{2}$$

Worked Solutions **Complex Numbers**

Q169 *MCQ*

For the expression State the complex conjugate of $4-7i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $4+7i$.

Correct option: D

Final answer

$4+7i$

Q170 *MCQ*

For the expression State the complex conjugate of $\sqrt{10}+2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{10}-2i$.

Correct option: A

Final answer

$\sqrt{10}-2i$

Worked Solutions **Complex Numbers**

Q171 MCQ

For the expression State the complex conjugate of $-8i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $8i$.

Correct option: B

Final answer

$8i$

Q172 MCQ

For $(3+2i)/(3-2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{5}{13} + \frac{12i}{13}$.

Correct option: B

Final answer

$$\frac{5}{13} + \frac{12i}{13}$$

Worked Solutions **Complex Numbers**

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{5}$.

Correct option: B

Final answer

$i\sqrt{5}$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{5}$.

Correct option: A

Final answer

$2i\sqrt{5}$

Worked Solutions **Complex Numbers**

Q175 **MCQ**

For $-\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i\sqrt{2}$.

Correct option: A

Final answer

$-3i\sqrt{2}$

Q176 **MCQ**

For $squarerootsof - 8$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{2}$.

Correct option: C

Final answer

$\pm 2i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 3.

Correct option: A

Final answer

3

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -4.

Correct option: B

Final answer

-4

Worked Solutions **Complex Numbers**

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: C

Final answer

$-2i$

Q180 MCQ

For $(2+\sqrt{-3})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $1+4i\sqrt{3}$.

Correct option: D

Final answer

$1+4i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q181 **MCQ**

For $\sqrt{-72}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: D

Final answer

$6i\sqrt{2}$

Q182 **MCQ**

For $squarerootsof-50$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 5i\sqrt{2}$.

Correct option: D

Final answer

$\pm 5i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6.

Correct option: A

Final answer

-6

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: D

Final answer

$-2i$

Worked Solutions **Complex Numbers**

Q185 MCQ

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: B

Final answer

$6i\sqrt{2}$

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $4i\sqrt{2}$.

Correct option: B

Final answer

$4i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q187 MCQ

For $(\sqrt{5}+\sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $3+2i\sqrt{10}$.

Correct option: A

Final answer

$3+2i\sqrt{10}$

Q188 MCQ

For $(3+\sqrt{-5})(3-\sqrt{-5})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 14.

Correct option: D

Final answer

14

Worked Solutions **Complex Numbers**

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{7}$.

Correct option: C

Final answer

$i\sqrt{7}$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{7}$.

Correct option: A

Final answer

$2i\sqrt{7}$

Worked Solutions **Complex Numbers**

Q191 MCQ

For $squarerootsof - 12$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{3}$.

Correct option: A

Final answer

$\pm 2i\sqrt{3}$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i$.

Correct option: A

Final answer

$-3i$

Worked Solutions **Complex Numbers**

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6.

Correct option: D

Final answer

-6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-4i$.

Correct option: C

Final answer

$-4i$

Worked Solutions **Complex Numbers**

Q195 MCQ

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{2}$.

Correct option: C

Final answer

$5i\sqrt{2}$

Q196 MCQ

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-1 + 2i\sqrt{2}$.

Correct option: A

Final answer

$-1 + 2i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q197 *MCQ*

For $(4+\sqrt{-3})(4-\sqrt{-3})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 19.

Correct option: B

Final answer

19

Q198 *MCQ*

For $\sqrt{-75}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{3}$.

Correct option: D

Final answer

$5i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q199 MCQ

Solve for the real values x and y: $(1+3i)(x+i)=x(y+i)$**:**

Solution

1. Expand and equate real and imaginary parts.
2. The system gives $(x=-1/2)$ and $(y=7)$.

Correct option: B

Final answer

$\left(-\frac{1}{2},\ 7\right)$

Q200 MCQ

Given x is real, solve: $(4x-4i)(x^2+xi)=10x^2i^{32}$**:**

Solution

1. Use $(i^{32}=1)$ and $((x-i)(x+i)=x^2+1)$.
2. The real equation is $(4x(x^2+1)=10x^2)$, giving $x=0,\ 1/2$, or 2.

Correct option: D

Final answer

$\left\{0,\ 2,\ \frac{1}{2}\right\}$

Worked Solutions **Complex Numbers**

Q201 MCQ

Simplify exactly: $(i(8-6i)/((1+2i)(3-i)))^{2027}$:

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The exact base simplifies to ((7+i)/5). Keep the 2027th power unexpanded.

Correct option: B

Final answer

$((7+i)/5)^{2027}$

Q202 MCQ

Solve over the complex numbers: $(3x-6)^2 = -8$:

Solution

1. Take square roots: ($3x - 6 = pm$ 2isqrt2).
2. Divide by 3 to obtain ($x = 2pm$

rac23isqrt2).

Correct option: B

Final answer

$x = 2 \pm (2/3)i\sqrt{2}$

Worked Solutions **Complex Numbers**

Quiz MCQ 1 **Concept Quiz · MCQ**

For 4-3i, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4 - 3i$ using the relevant family rule.
2. The correct value is $(4,\ -3)$.

Correct option: A

Final answer

$(4,\ -3)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For (1+2i)(2-i), select the simplified value:

Solution

1. Simplify $(1 + 2i)(2 - i)$ using the relevant family rule.
2. The correct value is $4 + 3i$.

Correct option: A

Final answer

$4 + 3i$

Worked Solutions **Complex Numbers**

Quiz MCQ 3 Concept Quiz · MCQ

For (3+i)/(1-i), select the value in the form a+bi:

Solution

1. Simplify $(3+i)/(1-i)$ using the relevant family rule.
2. The correct value is $1+2i$.

Correct option: D

Final answer

$1+2i$

Quiz MCQ 4 Concept Quiz · MCQ

For sqrt(-27), select the expression in terms of i:

Solution

1. Simplify $\sqrt{-27}$ using the relevant family rule.
2. The correct value is $3i\sqrt{3}$.

Correct option: A

Final answer

$3i\sqrt{3}$

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $(5/2, -\sqrt{5}/2)$

Final answer

$(5/2, -\sqrt{5}/2)$

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x,y: $(2+i)x + (1-i)y = 3 + i$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 4/3, y = 1/3$

Final answer

$x = 4/3, y = 1/3$

Worked Solutions **Complex Numbers**

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: -1

Final answer

-1

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x-3)^2 = -12$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 3 \pm 2i\sqrt{3}$

Final answer

$x = 3 \pm 2i\sqrt{3}$

Answer key

Use the question number shown on each page.

Q001 $(3,\ -2)$

Q002 $(4/3, -\sqrt{5}/3)$

Q003 $(-8,\ 0)$

Q004 $(0,\ 2)$

Q005 $x = -1, y = 4$

Q006 $(-3,\ 5)$

Q007 $(-1/2, -\sqrt{3}/2)$

Q008 $(1,\ 0)$

Q009 $(0,\ 3)$

Q010 $x = 6, y = -2$

Q011 $x = -2, y = 3$

Q012 $(2,\ 5)$

Q013 $(-25,\ 13)$

Q014 $(2/3, -\sqrt{5}/3)$

Q015 $(7/2, -\sqrt{2}/2)$

Q016 $(-4,\ 0)$

Q017 $(1/4,\ 0)$

Q018 $(0, \sqrt{3})$

Q019 $(0,\ -1)$

Q020 $x = 3, y = 2$

Q021 $x = 3, y = -3$

Q022 $x = 2, y = -2$

Q023 $x = 2, y = 1$

Q024 $x = 1/2, y = 3/2$

Q025 $x = -1, y = 3$

Q026 $1 - 7i$

Q027 $3 - 4i$

Q028 -1

Q029 $1 + 8i$

Q030 $8 - i$

Q031 0

Q032 $4 - i$

Q033 $1 - 3i$

Q034 -2

Q035 $-1 + 4\sqrt{2}i$

Q036 $7 + 11i$

Q037 $1/15 - 13/14i$

Q038 $-3 - 4i$

Q039 -3

Q040 4

Q041 i

Q042 -1

Q043 $-i$

Q044 $x = 0$ or $x = 1$

Q045 $2 - 3i$

Q046 $7 + i\sqrt{11}$

Q047 6

Q048 $5i$

Q049 $3/5 + 4/5i$

Q050 $-1/2 - 3/2i$

Q051 $1 + 4i$

Q052 $\sqrt{2} - i\sqrt{10}$

Q053 2

Q054 $8i$

Q055 $1 - i$

Q056 $-1 - i$

Q057 $3 - 4i$

Q058 $9 - 8i$

Q059 $8 - i\sqrt{3}$

Q060 5

Q061 -15

Q062 $\sqrt{7}$

Q063 $-i$

Q064 $50i$

Q065 $-i\sqrt{6}$

Q066 i

Q067 $7/5 - 1/5i$

Q068 $2/3 + \sqrt{5}/3i$

Q069 $2 + i$

Q070 $3/5 + 4/5i$

Q071 $9/13 - 7/13i$

Q072 $-i$

Q073 $1/3 - 2/3i$

Q074 $((-2 - 144i)/85)^{2027}$

Q075 $\pm i\sqrt{6}$

Q076 $-6\sqrt{2}$

Q077 $-(2/3)i$

Q078 $i\sqrt{7}$

Q079 $3i\sqrt{3}$

Q080 $-2i\sqrt{3}$

Q081 $\pm 2i\sqrt{3}$

Q082 $-3\sqrt{2}$

Q083 $-3i$

Q084 $7 + 6\sqrt{2}i$

Q085 $i\sqrt{2}$

Q086 $3i$

Q087 $3i\sqrt{5}$

Q088 $-2i\sqrt{6}$

Q089 $-6i$

Q090 $\pm i\sqrt{2}$

Q091 $\pm 3i\sqrt{2}$

Q092 $\pm 4i$

Q093 $-2\sqrt{3}$

Q094 $-14\sqrt{5}$

Q095 $-2i$

Q096 $-(3/2)i$

Q097 $13i\sqrt{2}$

Q098 $6i\sqrt{2}$

Q099 $-2 + 2i\sqrt{15}$

Q100 7

Q101 $x = 4 \pm (\sqrt{10}/2)i$

Q102 C

Q103 D

Q104 D

Q105 B
Q106 D
Q107 A
Q108 B
Q109 C
Q110 B
Q111 D
Q112 D
Q113 C
Q114 D
Q115 B
Q116 B
Q117 C
Q118 D
Q119 A
Q120 B
Q121 B
Q122 C
Q123 D
Q124 D
Q125 C
Q126 C
Q127 C
Q128 D
Q129 A
Q130 C
Q131 A
Q132 B
Q133 B
Q134 B
Q135 C
Q136 A
Q137 C
Q138 C
Q139 D
Q140 D
Q141 D
Q142 D
Q143 C
Q144 C
Q145 A
Q146 C
Q147 C
Q148 B
Q149 A
Q150 A
Q151 B
Q152 C
Q153 D
Q154 D
Q155 B
Q156 B
Q157 D
Q158 C
Q159 B
Q160 D
Q161 B
Q162 C
Q163 A
Q164 D
Q165 A
Q166 A
Q167 C
Q168 A
Q169 D
Q170 A
Q171 B
Q172 B
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Q175 A
Q176 C
Q177 A
Q178 B
Q179 C
Q180 D
Q181 D
Q182 D
Q183 A
Q184 D
Q185 B
Q186 B
Q187 A
Q188 D
Q189 C
Q190 A
Q191 A
Q192 A
Q193 D
Q194 C
Q195 C
Q196 A
Q197 B
Q198 D
Q199 B
Q200 D
Q201 B
Q202 B
Quiz MCQ 1 A
Quiz MCQ 2 A
Quiz MCQ 3 D
Quiz MCQ 4 A
Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Quiz WRITTEN 7 -1
Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$